

Modul 06 Basis Data
PRAKTIKUM BASIS DATA
Scripting Subquery C Nested Subquery



DISUSUN OLEH:
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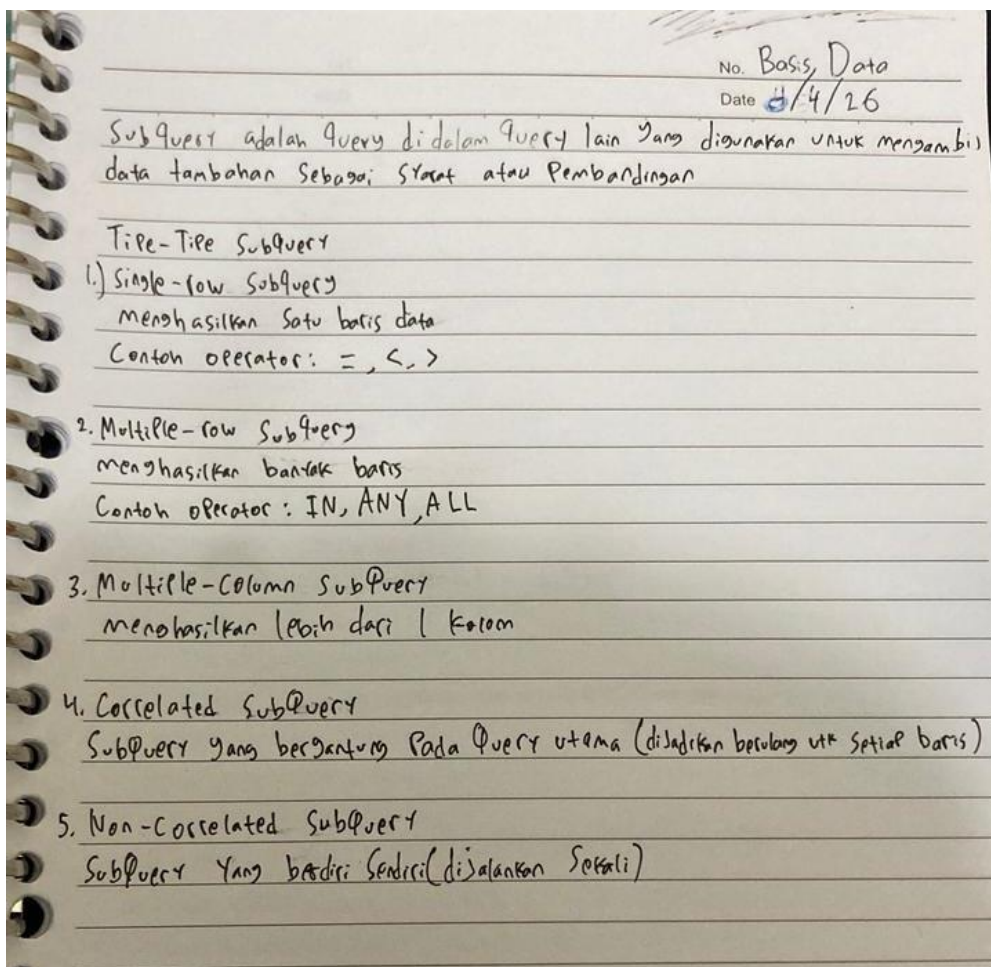
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FACULTY OF INFORMATICS
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TP

SOAL 1

Jelaskan definisi dari *subquery*? Serta sebutkan dan jelaskan tipe-tipe dari *subquery* (Tulis Tangan)!

JAWABAN



SOAL 02

Sebutkan hal-hal yang harus diperhatikan dalam melakukan proses subquery? (Minimal 5)

JAWABAN

Berikut hal-hal yang perlu diperhatikan dalam subquery:

1. Kesesuaian jumlah kolom

Hasil subquery harus sesuai dengan yang dibutuhkan query utama.

2. Kesesuaian tipe data

Tipe data hasil subquery harus sama dengan kolom pembanding.

3. Penggunaan operator yang tepat

Gunakan =, IN, ANY, ALL sesuai dengan jumlah hasil subquery.

4. Subquery harus di dalam tanda kurung ()

5. Performa query

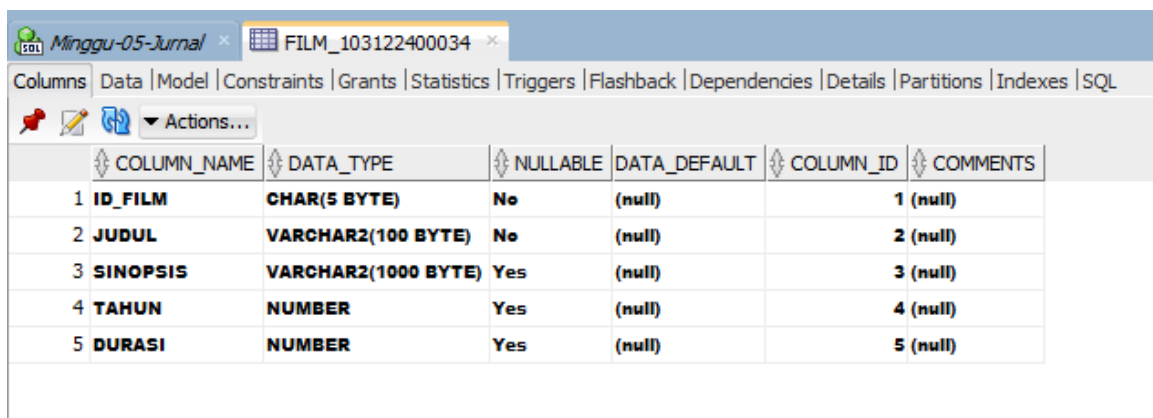
Subquery (terutama correlated) bisa lambat jika data besar.

6. Hasil subquery tidak boleh ambigu

Pastikan tidak menghasilkan lebih dari satu baris jika memakai operator tunggal (=).

3. Pastikan Nama setiap tabel berformat [Nama_Tabel][NIM] (SS BUKTI TABEL)

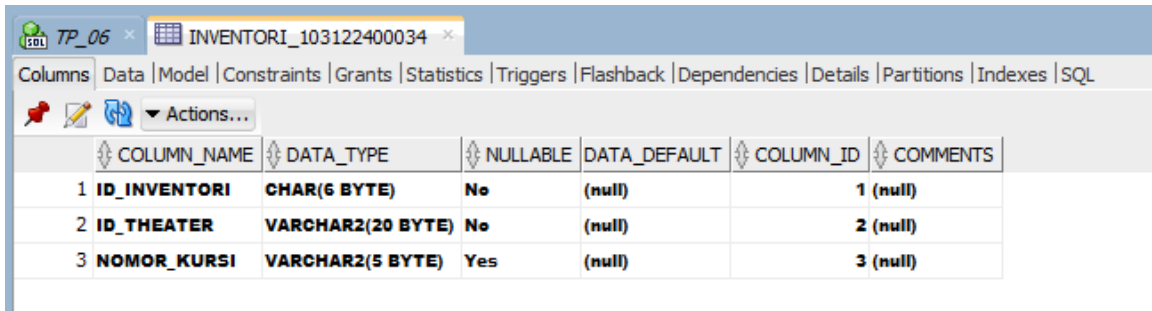
Tabel FILM



The screenshot shows the Oracle SQL Developer interface with the table structure of FILM_103122400034. The table has five columns: ID_FILM, JUDUL, SINOPSIS, TAHUN, and DURASI. The columns are defined with their respective data types, nullability, and default values.

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ID_FILM	CHAR(5 BYTE)	No	(null)	1	(null)
2	JUDUL	VARCHAR2(100 BYTE)	No	(null)	2	(null)
3	SINOPSIS	VARCHAR2(1000 BYTE)	Yes	(null)	3	(null)
4	TAHUN	NUMBER	Yes	(null)	4	(null)
5	DURASI	NUMBER	Yes	(null)	5	(null)

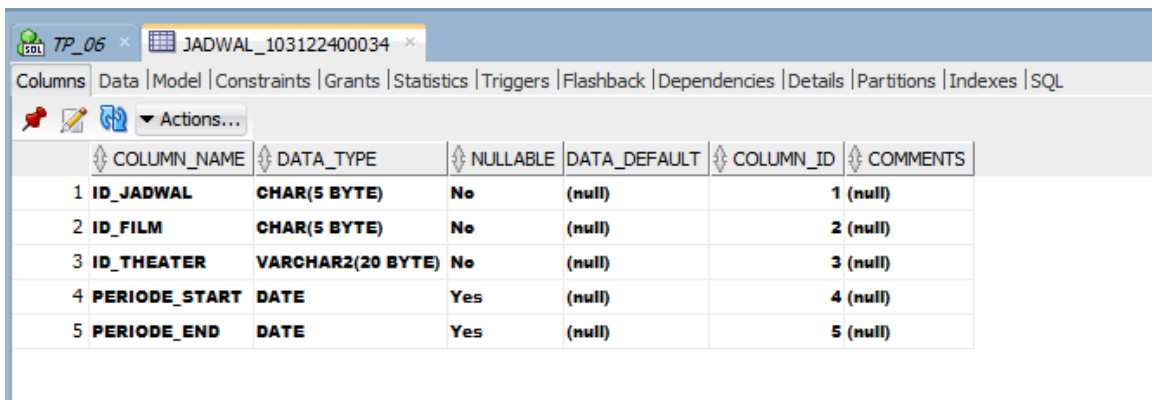
Tabel Inventory



The screenshot shows the SQL Developer interface for the table INVENTORI_103122400034. The 'Columns' tab is selected, displaying a table with 6 columns: COLUMN_NAME, DATA_TYPE, NULLABLE, DATA_DEFAULT, COLUMN_ID, and COMMENTS. There are 3 rows of data.

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ID_INVENTORI	CHAR(6 BYTE)	No	(null)	1	(null)
2	ID_THEATER	VARCHAR2(20 BYTE)	No	(null)	2	(null)
3	NOMOR_KURSI	VARCHAR2(5 BYTE)	Yes	(null)	3	(null)

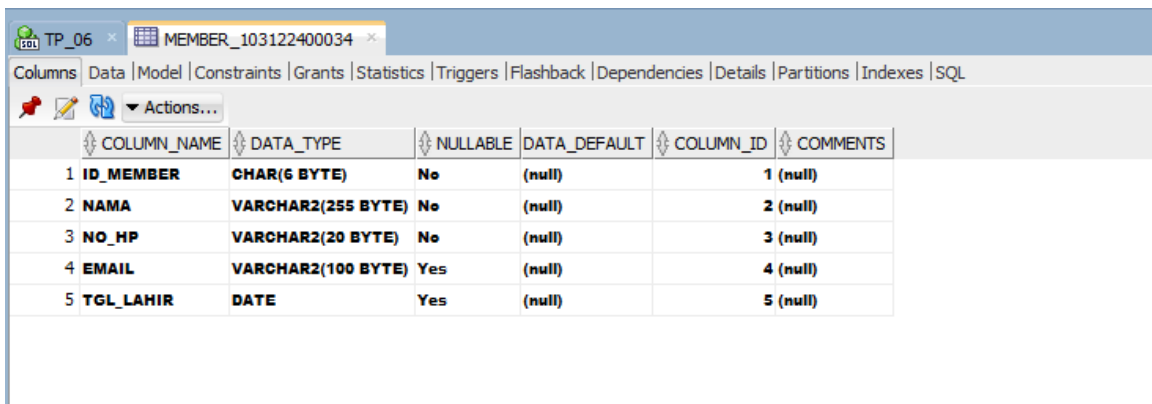
Tabel Jadwal



The screenshot shows the SQL Developer interface for the table JADWAL_103122400034. The 'Columns' tab is selected, displaying a table with 6 columns: COLUMN_NAME, DATA_TYPE, NULLABLE, DATA_DEFAULT, COLUMN_ID, and COMMENTS. There are 5 rows of data.

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ID_JADWAL	CHAR(5 BYTE)	No	(null)	1	(null)
2	ID_FILM	CHAR(5 BYTE)	No	(null)	2	(null)
3	ID_THEATER	VARCHAR2(20 BYTE)	No	(null)	3	(null)
4	PERIODE_START	DATE	Yes	(null)	4	(null)
5	PERIODE_END	DATE	Yes	(null)	5	(null)

Tabel Member



The screenshot shows the SQL Developer interface for the table MEMBER_103122400034. The 'Columns' tab is selected, displaying a table with 6 columns: COLUMN_NAME, DATA_TYPE, NULLABLE, DATA_DEFAULT, COLUMN_ID, and COMMENTS. There are 5 rows of data.

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ID_MEMBER	CHAR(6 BYTE)	No	(null)	1	(null)
2	NAMA	VARCHAR2(255 BYTE)	No	(null)	2	(null)
3	NO_HP	VARCHAR2(20 BYTE)	No	(null)	3	(null)
4	EMAIL	VARCHAR2(100 BYTE)	Yes	(null)	4	(null)
5	TGL_LAHIR	DATE	Yes	(null)	5	(null)

Tabel Theater

TP_06

THEATER_103122400034

Columns | Data | Model | Constraints | Grants | Statistics | Triggers | Flashback | Dependencies | Details | Partitions | Indexes | SQL

Actions...

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ID_THEATER	VARCHAR2(20 BYTE)	No	(null)	1 (null)	
2	KELAS	VARCHAR2(20 BYTE)	Yes	(null)	2 (null)	
3	HARGA	NUMBER	No	(null)	3 (null)	
4	KAPASITAS	NUMBER	Yes	(null)	4 (null)	

Tabel Transaksi

TP_06

TRANSAKSI_103122400034

Columns

Data

Model

Constraints

Grants

Statistics

Triggers

Flashback

Dependencies

Details

Partitions

Indexes

SQL

Actions...

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	KODE_PEMESANAN	CHAR(6 BYTE)	No	(null)	1 (null)	
2	ID_MEMBER	CHAR(6 BYTE)	No	(null)	2 (null)	
3	ID_JADWAL	CHAR(5 BYTE)	No	(null)	3 (null)	
4	ID_INVENTORI	CHAR(6 BYTE)	No	(null)	4 (null)	
5	TANGGAL	DATE	No	(null)	5 (null)	
6	STATUS	VARCHAR2(20 BYTE)	No	(null)	6 (null)	

4. Buatlah subquery untuk menampilkan judul film dengan durasi terpendek dan umur termuda yang menonton film tersebut

Query

```

SELECT
    F.JUDUL ||
    ' adalah film dengan durasi terpendek yaitu ' ||
    F.DURASI ||
    ' Menit, yang ditonton dengan umur termuda, yaitu ' ||
    (SELECT MIN(FLOOR(MONTHS_BETWEEN(SYSDATE, M.TGL_LAHIR)/12))
    FROM MEMBER_103122400034 M
    JOIN TRANSAKSI_103122400034 T

```

```

        ON M.ID_MEMBER = T.ID_MEMBER

JOIN JADWAL_103122400034 J

        ON T.ID_JADWAL = J.ID_JADWAL

WHERE J.ID_FILM = F.ID_FILM)

|| ' Tahun' AS KETERANGAN

FROM FILM_103122400034 F

WHERE F.DURASI = (

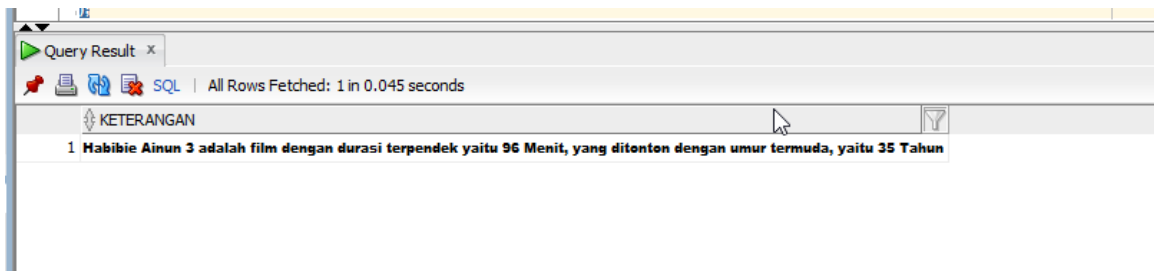
    SELECT MIN(DURASI)

    FROM FILM_103122400034

);

```

OUTPUT



5. Buatlah subquery untuk menampilkan nomor teater, kelas, dan total pemakaian teater tersebut yang paling sering digunakan dengan menggunakan join on

Query

```

SELECT TH.ID_THEATER,

        TH.KELAS,

        COUNT(J.ID_THEATER) AS TOTAL_PEMAKAIAN

FROM THEATER_103122400034 TH

JOIN JADWAL_103122400034 J

```

```

ON TH.ID_THEATER = J.ID_THEATER

GROUP BY TH.ID_THEATER, TH.KELAS

HAVING COUNT(J.ID_THEATER) = (

    SELECT MAX(total_pakai)

    FROM (

        SELECT COUNT(ID_THEATER) AS total_pakai

        FROM JADWAL_103122400034

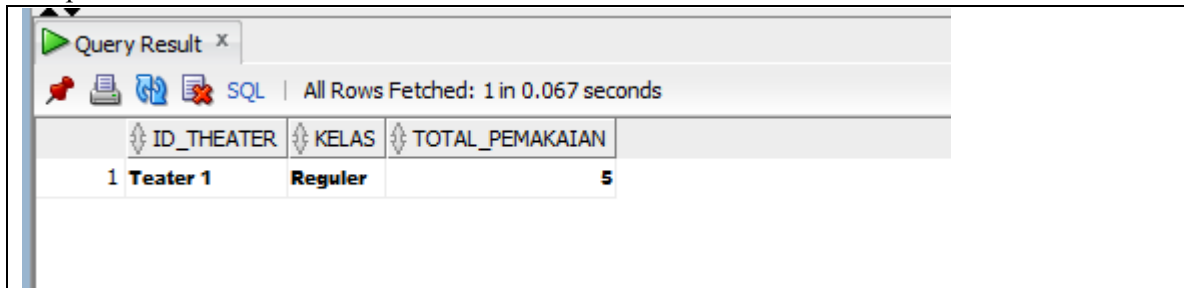
        GROUP BY ID_THEATER

    )

);

```

Output



The screenshot shows a 'Query Result' window with a table containing one row. The table has three columns: ID_THEATER, KELAS, and TOTAL_PEMAKAIAN. The first row contains the values '1 Teater 1', 'Regular', and '5'.

ID_THEATER	KELAS	TOTAL_PEMAKAIAN
1 Teater 1	Regular	5

